

The Impact of White-Box Adversarial Attacks on Group Fairness in Machine Learning Models for Prostate Cancer Diagnosis

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Abstract—AI systems deployed in healthcare institutions must be both robust to adversarial attacks and fair across different subgroups. This work investigates the impact of white-box adversarial attacks on group fairness using the patients’ age as the sensitive attribute. We evaluated three machine learning models under three white-box adversarial attacks, and we measured group fairness using three group fairness metrics. Results from both cross-validation and the holdout test dataset showed that group fairness is improved under adversarial conditions. As adversarial attacks reduce the models’ performance, the selection rate difference and equalized odds difference show statistically significant improvements across all models. However, these improvements reflect a reduction in the models’ discriminative ability rather than actual group fairness improvements that could have been learned during training. Furthermore, the results showed that the support vector machine maintained a stable recall difference under adversarial conditions, which suggests that support vectors can preserve the true positive rate across different groups. From a clinical perspective, the results from this work suggest that healthcare institutions deploying AI diagnostic systems should continuously monitor both predictive accuracy and group fairness, as unexpected improvements in group fairness may indicate that the system’s performance has been reduced.

Index Terms—adversarial robustness, algorithmic fairness, trustworthy AI, prostate cancer, radiomics

I. INTRODUCTION

Machine Learning (ML) and Deep Learning (DL) models have significantly progressed the healthcare domain due to their ability to provide fast and accurate predictions that have

learned from large amounts of data. Such accuracy is important in critical clinical operations because clinicians can improve their decision-making process and provide faster and more accurate treatment plans to their patients. For instance, in medical diagnosis, ML models are widely used to automate the detection and classification of various diseases from various data modalities, such as radiomics features, which are extracted from medical images [1], [2].

With the increasing involvement of automated tools powered by Artificial Intelligence (AI), ensuring that the generated predictions are fair is essential, as any source of bias could have a negative impact on the patients who belong to specific groups. Individual fairness in AI is achieved when AI generated predictions are similar for individuals who belong to different subgroups but share similar characteristics, such as their race, gender, or age. Similarly, group fairness in AI is achieved when the AI system produces an equal proportion of outcomes for all the subgroups [3], [4], [5].

However, even when an AI system has been generalized to the training data distribution and, as a result, generates accurate predictions under normal conditions, its performance during inference can be reduced by the addition of small, imperceptible-to-the-human-eye adversarial noise to the original input, which leads to the reduction of the trustworthiness and reliability of its performance [6], [7]. For example, if adversarial attacks manipulate the data distribution during inference, there is a high probability that the AI system will

generate false negatives (FNs) more frequently, which could put patients' lives in danger.

Research in adversarial robustness focuses on understanding the impact of the adversarial attacks on AI models' performance [6], [8], while research in AI fairness focuses on understanding how much the AI generated predictions discriminate across different sensitive groups [3], [4]. Furthermore, existing research in the intersection of these fields has been mainly focused on measuring bias under data poisoning attacks [9]. In this category, an attacker intentionally manipulates the training data distribution to alter the model's predictions to discriminate against specific individuals or sensitive groups [10]. Unlike prior works that primarily evaluated fairness under data poisoning attacks, in this work, we address this gap by evaluating the impact of white-box adversarial attacks on group fairness.

From a clinical perspective, AI diagnostic systems that are deployed in healthcare are vulnerable to inference-time adversarial attacks, which could lead them to generate wrong predictions. Unlike data poisoning attacks that target the training process before the deployment, white-box adversarial attacks target the systems that have already been deployed in the real world. The objective of this work is to investigate how white-box adversarial attacks affect group fairness across patients of different age groups in prostate cancer diagnosis. The experimental results revealed a phenomenon that we call *fairness through performance degradation*, where adversarial attacks paradoxically improve group fairness. This effect occurs not because the models have been trained to behave fairly across the different sensitive groups, but because their discriminative ability has been dropped due to the reduction of their predictive accuracy under adversarial conditions. Based on these findings, we recommend the continuous monitoring of both predictive accuracy and group fairness of AI systems that have already been in the real world, as unexpected fairness improvements may provide early evidence of performance degradation.

II. MATERIALS & METHODS

A. Background

1) *Adversarial Attacks*: They present serious security risks in AI systems, as they can degrade their predictive performance. These methods can be categorized based on the attacker's knowledge about the AI system. When all the knowledge is available, such as the AI system's weights, architecture, and training dataset, adversarial attacks are categorized as white box [10]. Conversely, when minimal knowledge is available, they are categorized as a black box [10]. In this work, we have focused on white-box attacks because adversarial examples generated from attacks belonging to this category are considered stronger compared to those generated by black-box attacks, as having complete knowledge of the underlying AI system can be considered a significant advantage.

The first adversarial attack used in our work is the Fast Gradient Sign Method (FGSM) [6], which is defined as follows:

$$x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x J(\theta, x, y)) \quad (1)$$

where (x, y) corresponds to the original input with its corresponding ground truth label, ϵ is a hyperparameter that corresponds to the magnitude of the attack, J represents the loss function used to train the AI model, and θ represents the model parameters, such as its weights. FGSM is considered a computationally efficient adversarial attack since it requires the calculation of only one optimization step in the gradient direction of the loss function with respect to the input.

The second adversarial attack we used is the Projected Gradient Descent (PGD) [8], which is defined as:

$$x_{t+1} = (x_t + \alpha \cdot \text{sign}(\nabla_x J(\theta, x_t, y))) \quad (2)$$

where x_t represents the adversarial example at the iteration step t , and α is the step size. PGD is an extended version of FGSM because it performs multiple optimization iterations in the gradient direction of the input x , resulting in stronger adversarial examples compared to those generated by FGSM.

The third adversarial attack used in this work is Carlini-Wagner (C&W) [11], which can be defined by solving the following optimization problem:

$$\min_{\delta} \|\delta\|_2^2 + c \cdot f(x + \delta) \quad (3)$$

where δ is the adversarial noise, c is a hyperparameter that balances δ with the attack success $f(x + \delta)$. C&W attack is considered a computationally expensive method because it calculates the optimal adversarial noise δ that will cause the model to misclassify the input x with high probability.

2) *Group Fairness Metrics*: They measure the levels of discrimination that an AI system has in its predictions across different demographic groups, which are defined based on their sensitive attributes (e.g., race, gender, and age, among others). For an AI system to be considered fair, those attributes should not influence its predictions in ways that might disadvantage specific populations [4], [5].

For a binary classification problem containing true labels $Y \in \{0, 1\}$ and a population that can be distinguished into distinct groups, $G = \{g_1, g_2, \dots, g_k\}$ with the individuals in each group g containing sensitive characteristics, which can be denoted as $A = \{a_1, a_2, \dots, a_k\}$, group fairness requires that the predictions produced by the AI system remain fair across all groups $g \in G$. This means that the predictions made by an AI system should not be influenced by any of the groups' sensitive attribute $\alpha \in A$.

The selection rate difference (SR) measures whether the positive predictions made by the AI system are distributed equally across all the groups and can be defined as:

$$\Delta_{SR} = \max_{g \in G} \Pr(\hat{Y} = 1 | G = g) - \min_{g \in G} \Pr(\hat{Y} = 1 | G = g) \quad (4)$$

This metric provides useful information regarding the positive predictions generated by the AI system. High values indicate

that the AI system assigns the positive label ($\hat{Y} = 1$) to some subgroups more frequently compared to others.

Recall difference [5] provides information regarding the AI system's ability to assign an equal true positive rate (TPR) for each $g \in G$ and is defined as:

$$\Delta_{TPR} = \max_{g \in G} TPR_g - \min_{g \in G} TPR_g \quad (5)$$

where $TPR_g = \Pr(\hat{Y} = 1 | Y = 1, G = g)$ refers to the TPR for the subgroup g .

Equalized odds (EO) difference [5] ensures that the AI system assigns similar false positive rates (FPRs) and TPRs across subgroups and is defined as:

$$\Delta_{EO} = \max(\Delta_{TPR}, \Delta_{FPR}) \quad (6)$$

where $\Delta_{FPR} = \max_{g \in G} FPR_g - \min_{g \in G} FPR_g$ and $FPR_g = P(\hat{Y} = 1 | Y = 0, G = g)$ indicates the FPR for the group g .

B. Dataset

In this study, the Prostate-Net¹ dataset from the Procancer-I² project was used. This dataset contains radiomics features [12] from 6674 retrospectively and 1746 prospectively collected bpMRI data, including T2-weighted (T2w), diffusion-weighted imaging (DWI), and apparent diffusion coefficient maps (ADC). For the assessment of the models' group fairness under adversarial conditions, we used the patients' age as the sensitive attribute, as no other sensitive characteristics were available.

Data Preprocessing: In this work, we have developed a tool utilizing the Python programming language that automates the data preprocessing pipeline for datasets containing tabular data, which we made publicly available at GitHub³.

C. Models

We have conducted the group fairness evaluation under three white-box adversarial attacks described in Section II-A1 using the following AI models: (i) Logistic Regression (LR), which is widely used in supervised learning classification problems to predict the probability of a specific outcome using the logistic function [14]. (ii) Support Vector Machine (SVM), which classifies data by constructing an optimal decision boundary that maximizes the distance between data points of different classes [14], and (iii) Neural Network (NN), which consists of an input layer, one or more hidden layers with nonlinear activation functions, and an output layer with a sigmoid activation function for binary classification problems. The network learns by minimizing a loss function through backpropagation, adjusting its weights via gradient descent to construct the optimal decision boundary between the different classes existing in the dataset [14].

D. Experimental Design

The prostate cancer dataset, which is presented in Section II-B was split into a training dataset (70%, 6,825 patients), a validation dataset (15%, 1,462 patients), and a holdout test dataset (15%, 1,463 patients). Then, the following two evaluation phases were performed:

Cross-Validation: In the training dataset a 3-fold cross-validation was applied. Specifically, we used hyperparameter optimization to find the best architectures from each fold. For ML models, we used grid search, and for the NN, we used Neural Architecture Search (NAS). The adversarial attacks were then used on each validation fold. Group Fairness metrics were calculated before and after the attacks to observe the group fairness levels of the models' predictions across the patients belonging to different aged groups. The results presented in Table Ia and Table Ib show the average results for the test accuracy and group fairness metrics with the standard error of the mean (SEM) across the validation folds under normal and adversarial conditions.

Holdout Test Dataset: In this phase, the models were trained on the entire 70% training dataset. Similarly, the best architectures for all the models were found using the hyperparameter optimization strategy described in the cross-validation phase. Then, the adversarial attacks were applied on the 15% holdout test dataset. Table Ia and Table Ib present the average results for the test accuracy and group fairness metrics with the standard error of the mean (SEM) under normal and adversarial conditions.

III. RESULTS

Table Ia and Table Ib present the average results for the test accuracy and group fairness metrics under normal and adversarial conditions, across the validation folds and holdout test dataset. To determine the statistical significance of the experimental results, a paired t-test for the cross-validation results and a one-sample t-test on the holdout test dataset results were performed. Furthermore, due to the nine comparisons between the three models and three group fairness metrics, we applied the Bonferroni correction [15], a method that provides a probability threshold to control the occurrence of false positives (FPs) that appear during the statistical analysis.

According to Table Ia, the test accuracy significantly decreased under adversarial conditions in both evaluation phases. This decline in performance indicates the high vulnerability levels of the models to the noise generated by adversarial attacks. Table Ib demonstrates that adversarial attacks reduced both the selection rate difference and equalized odds difference, with these reductions being statistically significant. This finding suggests that adversarial attacks produced models with improved group fairness. This means that the models treated patients belonging to different age groups equally compared to normal conditions. Similarly, the significant reduction in the recall difference for both the LR and NN models suggests that, in the presence of adversarial noise, these two models performed fairly among patients with clinically significant

¹<https://prostatenet.eu/>

²<https://www.procancer-i.eu/>

³https://github.com/KosmasApostolidis/automated_tabular_preprocessor

TABLE I: Test Accuracy and Group Fairness Metrics Before and After Adversarial Attacks

(a) Test Accuracy (mean $\pm$ SEM)

Model	Cross-Validation		Holdout Test Dataset	
	Before	After	Before	After
LR	0.79 $\pm$ 0.001	0.36 $\pm$ 0.02	0.77	0.36 $\pm$ 0.03
SVM	0.8 $\pm$ 0.001	0.54 $\pm$ 0.02	0.79	0.55 $\pm$ 0.03
NN	0.82 $\pm$ 0.002	0.36 $\pm$ 0.02	0.81	0.33 $\pm$ 0.02

(b) Group Fairness Metrics (mean $\pm$ SEM)

Model	Metric	Cross-Validation			Holdout Test Dataset		
		Before	After	<i>p</i> value	Before	After	<i>p</i> value
LR	Δ_{SR}	0.41 $\pm$ 0.01	0.21 $\pm$ 0.02	0.00005	0.46	0.22 $\pm$ 0.05	0.001
	Δ_{Rec}	0.23 $\pm$ 0.02	0.15 $\pm$ 0.01	0.0002	0.32	0.14 $\pm$ 0.03	0.0004
	Δ_{EO}	0.33 $\pm$ 0.02	0.23 $\pm$ 0.02	0.0005	0.41	0.26 $\pm$ 0.03	0.001
SVM	Δ_{SR}	0.25 $\pm$ 0.01	0.09 $\pm$ 0.01	0.00003	0.24	0.12 $\pm$ 0.03	0.002
	Δ_{Rec}	0.1 $\pm$ 0.01	0.09 $\pm$ 0.01	0.63	0.1	0.12 $\pm$ 0.02	0.254
	Δ_{EO}	0.31 $\pm$ 0.02	0.13 $\pm$ 0.01	0.0001	0.25	0.14 $\pm$ 0.03	0.002
NN	Δ_{SR}	0.37 $\pm$ 0.01	0.17 $\pm$ 0.02	0.000003	0.39	0.22 $\pm$ 0.03	0.004
	Δ_{Rec}	0.2 $\pm$ 0.02	0.11 $\pm$ 0.01	0.0003	0.2	0.09 $\pm$ 0.01	0.00005
	Δ_{EO}	0.32 $\pm$ 0.02	0.21 $\pm$ 0.02	0.0007	0.4	0.32 $\pm$ 0.02	0.005

Statistical Significance: Cross-Validation (Paired t-test), Holdout (One-sample t-test), $\alpha = 0.0056$, Bold *p*-values = non-significant. "Before" values in Holdout Test Dataset lack SEM because they represent single observations.

prostate cancer. In contrast, the SVM model provided statistically insignificant results for both the cross-validation ($p_{\text{value}} = 0.63$) and the holdout test dataset ($p_{\text{value}} = 0.254$) regarding the recall difference, indicating that adversarial noise did not affect SVM's true positive rate compared to the other models.

IV. DISCUSSION

In this work, we have conducted a group fairness evaluation of three ML models under three white-box adversarial attacks on a prostate cancer dataset that contains radiomics features from 9,750 patients using their age as the sensitive attribute.

The test accuracy results from Table Ia showed that the performance of the models is drastically reduced under adversarial conditions. Specifically, the performance from all models was consistently lower on both the cross-validation and the holdout test dataset under adversarial conditions compared to normal conditions. This shows that adversarial attacks successfully reduce the performance of AI models and make them generate wrong predictions [6], [7]. However, based on the experimental results from Table Ia, the SVM model achieved the highest accuracy under adversarial conditions compared to LR and NN, leading to a stable recall difference.

Moreover, the results from Table Ib revealed that there exists a phenomenon where adversarial attacks "improve group fairness" while reducing the test accuracy at the same time [9]. The selection rate difference and equalized odds difference decreased with statistical significance across all models under adversarial conditions, indicating that the models behaved more fairly across patients with different age groups. These improvements may seem like actual improvements in fairness, but they are actually an example of *fairness through performance degradation*, when the models behave fairly, but in reality, they aren't fair at all. This happens not because the

models were trained to be fair, but because they have lost their discrimination ability due to their performance reduction under the influence of adversarial attacks.

On the other hand, the experiments showed that SVM produced statistically insignificant results under adversarial conditions for the recall difference across both evaluation phases ($p_{\text{value}} = 0.63$ for the cross-validation and $p_{\text{value}} = 0.254$ for the holdout test dataset) compared to LR and NN ($p_{\text{value}} < 0.0056$ in both phases). Specifically, Table Ia shows that under adversarial conditions, SVM achieved the highest accuracy compared to LR and NN, which means that its decision boundary was less affected. Also, the decision boundary of SVM only depends on support vectors. If adversarial attacks do not drastically change the position of the support vectors, then the true positive rate remains stable. This indicates that margin-based classifiers demonstrate greater adversarial robustness compared to gradient-based classifiers, such as neural networks [16].

From a clinical perspective, the results from this work have direct implications for AI systems used for prostate cancer diagnosis. First, the fairness through performance degradation phenomenon suggests that unexpected improvements in group fairness should be considered as a potential indicator of the AI system's degradation in its predictive accuracy. Second, in healthcare, it is important to ensure that the difference in true positive rates between different sensitive groups remains the same [3]. If maintaining a stable recall difference is the primary concern, then incorporating the SVM architecture into an AI diagnostic system might be preferable compared to other architectures.

Several limitations have been recognized and left as future work. First, the experiments were performed on a single dataset with radiomics features, and the results may differ

if other data modalities, such as medical images, are used instead. Second, the group fairness evaluation is performed only on white-box attacks. On the other hand, if the group fairness evaluation is performed under black-box attacks, then the results observed in this work could be different.

V. CONCLUSION

This work investigates the impact of white-box adversarial attacks on the models' group fairness. The experimental results from both evaluation phases showed that while adversarial attacks reduce the models' predictive accuracy, they simultaneously lead to improvements in group fairness. As a result, these fairness improvements are achieved because the models have lost their discriminative ability under the influence of adversarial attacks, which leads the models to treat different sensitive groups more fairly compared to normal conditions. Future work should extend the current group fairness analysis on different data modalities and types of adversarial attacks.

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